



Last Resort Hotels

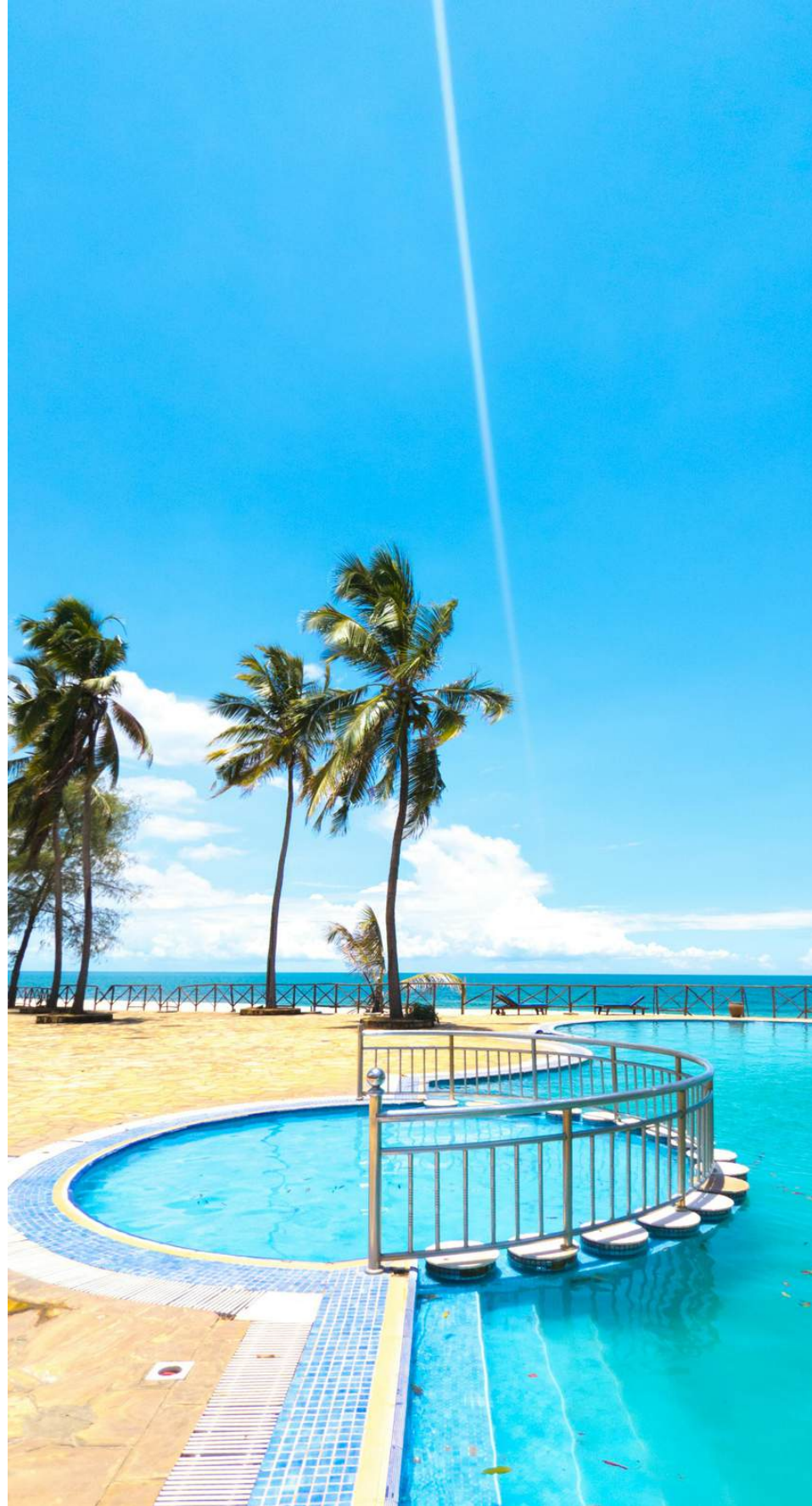
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Start Slide

Background

- Last Resort Hotels has expanded rapidly across multiple luxury properties
- Complex structures: buildings → wings → floors → rooms
- Supports both:
 - overnight guests (sleeping rooms)
 - event hosts (meeting rooms)
- Operations became difficult to manage manually





Core Problem

- Staff need real-time access for:
- Room availability and status
 - Guest stay information
 - Event scheduling and room usage
 - Billing and charge responsibility
 - Service usage (meals, spa, etc.)
 - Guest location and access tracking
- Without a system:
- Overbooking risk
 - Billing errors
 - Poor guest experience

System Solution Overview



Our system integrates:

- Database (SQLite)
- ERD (fully normalized design)
- SQL queries (data retrieval & analytics)
- Web interface (Flask)

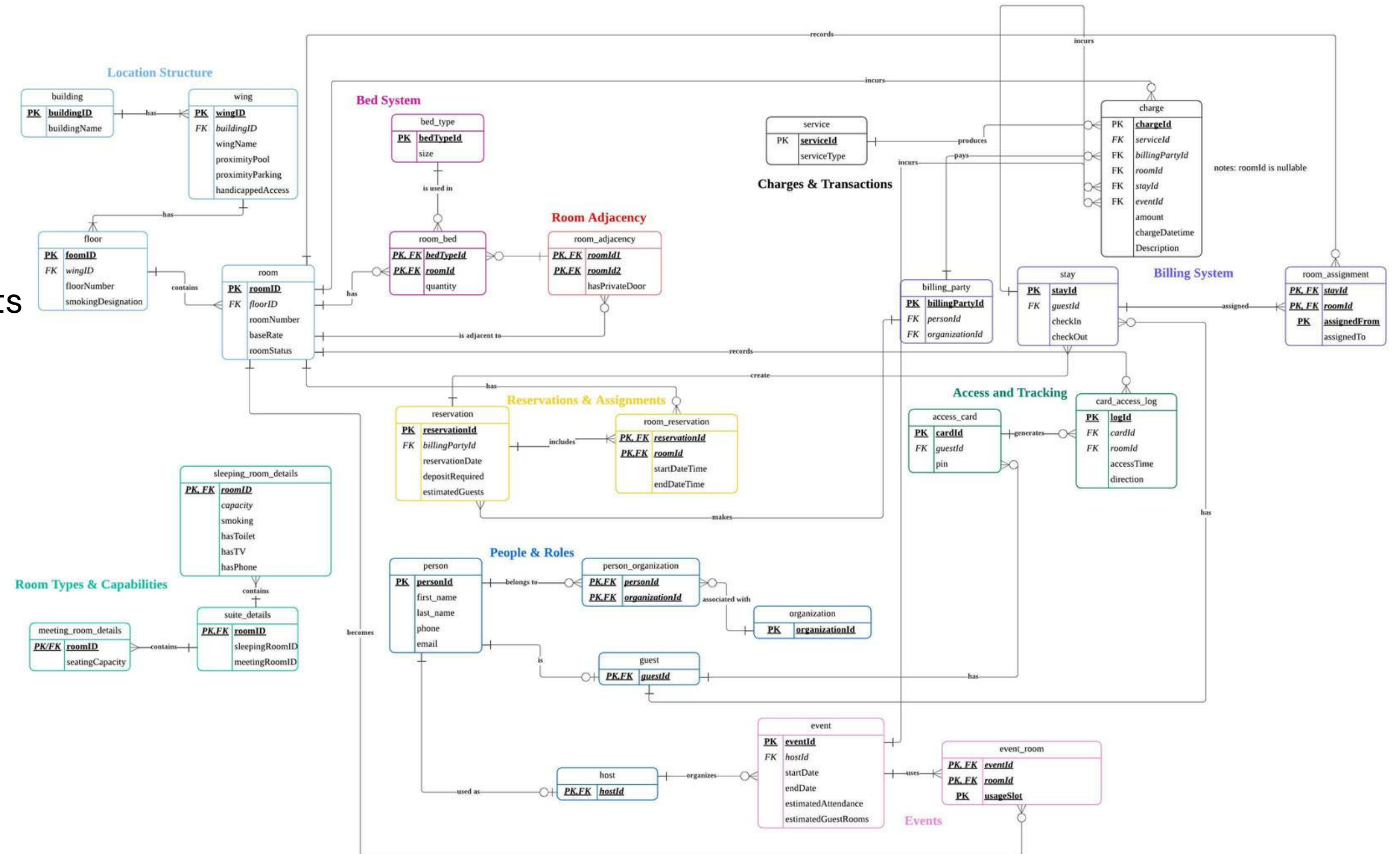
Supports:

- Operations management
- Real-time queries
- Staff interaction

ERD Overview

10 main modules:

- Location structure
- Bed System
- Room Adjacency
- Reservations & Assignments
- Room Types & Capabilities
- People & Roles
- Events
- Access and Tracking
- Charges & Transactions
- Billing System



Schema Design

- ERD translated into SQLite tables
- room stores room number, rate, status, and floor location
- bed_type standardizes bed size options
- room_bed connects rooms with bed types
- Composite primary key prevents duplicate room-bed records
- Foreign keys maintain relationships between tables
- CHECK (quantity > 0) ensures valid bed quantities

```
CREATE TABLE room (  
    roomID          INTEGER PRIMARY KEY,  
    floorID         INTEGER NOT NULL,  
    roomNumber      TEXT NOT NULL,  
    baseRate        REAL,  
    roomStatus      TEXT,  
    FOREIGN KEY (floorID) REFERENCES floor(floorID)  
);
```

```
CREATE TABLE bed_type (  
    bedTypeId       INTEGER PRIMARY KEY,  
    size            TEXT NOT NULL  
);
```

```
CREATE TABLE room_bed (  
    bedTypeId       INTEGER NOT NULL,  
    roomId          INTEGER NOT NULL,  
    quantity        INTEGER NOT NULL,  
    PRIMARY KEY (bedTypeId, roomId),  
    FOREIGN KEY (bedTypeId) REFERENCES bed_type(bedTypeId),  
    FOREIGN KEY (roomId) REFERENCES room(roomID),  
    CHECK (quantity > 0)  
);
```


Data Population

- Inserted sample hotel room data
- Added room numbers, rates, and statuses
- Created standard bed types
- Linked rooms to bed types through room_bed
- Supports realistic room configurations
- Provides base data for reservations and reports

```
INSERT INTO room_bed (bedTypeId, roomId, quantity) VALUES
(3, 101, 2),
(4, 102, 1),
(1, 103, 2),
(2, 104, 2),
(3, 105, 1),
(4, 106, 1),
(1, 107, 2),
(2, 108, 2),
(3, 109, 1),
(4, 110, 1),
```

```
INSERT INTO room (roomId, floorID, roomNumber, baseRate, roomStatus) VALUES
(101, 1, 'A101', 120.00, 'available'),
(102, 1, 'A102', 120.00, 'available'),
(103, 1, 'A103', 135.00, 'occupied'),
(104, 2, 'A201', 150.00, 'available'),
(105, 2, 'A202', 150.00, 'cleaning'),
(106, 2, 'A203', 165.00, 'available'),
(107, 3, 'B101', 110.00, 'available'),
(108, 3, 'B102', 110.00, 'occupied'),
(109, 4, 'B201', 145.00, 'available'),
```

```
INSERT INTO bed_type (bedTypeId, size) VALUES
(1, 'Twin'),
(2, 'Double'),
(3, 'Queen'),
(4, 'King'),
(5, 'Sofa Bed'),
(6, 'Rollaway');
```

Query Categories

```
-- Show stay history for each guest
SELECT s.guestId, p.first_name, p.last_name, s.checkIn, s.checkOut
FROM stay s
JOIN person p ON s.guestId = p.personId;

-- Show assigned rooms for each stay
SELECT ra.stayId, r.roomNumber, ra.assignedFrom, ra.assignedTo
FROM room_assignment ra
JOIN room r ON ra.roomId = r.roomID;

-- Show rooms used in each event
SELECT er.eventId, r.roomNumber, er.usageSlot
FROM event_room er
JOIN room r ON er.roomId = r.roomID;

-- Total charge per billing party
SELECT billingPartyId, ROUND(SUM(amount), 2) AS total
FROM charge
GROUP BY billingPartyId;

-- Revenue by service type
SELECT s.serviceType, ROUND(SUM(c.amount), 2) AS revenue
FROM charge c
JOIN service s ON c.serviceId = s.serviceId
GROUP BY s.serviceType;
```

```
-- Show full room hierarchy
SELECT r.roomID, r.roomNumber, r.baseRate,
       f.floorNumber, w.wingName, b.buildingName
FROM room r
JOIN floor f ON r.floorID = f.floorID
JOIN wing w ON f.wingID = w.wingID
JOIN building b ON w.buildingID = b.buildingID;

-- Show bed configuration of each room
SELECT r.roomNumber, bt.size, rb.quantity
FROM room_bed rb
JOIN room r ON rb.roomId = r.roomID
JOIN bed_type bt ON rb.bedTypeId = bt.bedTypeId;

-- Show rooms reserved in each reservation
SELECT rr.reservationId, r.roomNumber, rr.startDateTime, rr.endDateTime
FROM room_reservation rr
JOIN room r ON rr.roomId = r.roomID;

-- Reservations for a specific room
SELECT reservationId, startDateTime, endDateTime
FROM room_reservation
WHERE roomId = 101;
```

- Room & Location Queries
- Reservation & Availability Queries
- Guest & Stay Queries
- Events Queries
- Billing & Revenue Queries
- Access Tracking Queries

Application / Code

```
@app.route("/rooms")
def rooms():
    conn = get_db_connection()

    rooms = conn.execute("""
        SELECT r.roomID, r.roomNumber, r.baseRate, r.roomStatus,
               f.floorNumber, w.wingName, b.buildingName
        FROM room r
        JOIN floor f ON r.floorID = f.floorID
        JOIN wing w ON f.wingID = w.wingID
        JOIN building b ON w.buildingID = b.buildingID
        ORDER BY b.buildingName, w.wingName, f.floorNumber, r.roomNumber
    """).fetchall()

    conn.close()
    return render_template("rooms.html", rooms=rooms)

    if room is None:
        conn.close()
        return render_template("error.html", error="Room not found.")

    beds = conn.execute("""
        SELECT bt.size, rb.quantity
        FROM room_bed rb
        JOIN bed_type bt ON rb.bedTypeId = bt.bedTypeId
        WHERE rb.roomId = ?
        ORDER BY bt.size
    """, (room_id,)).fetchall()

adjacent_rooms = conn.execute("""
    SELECT r.roomNumber, ra.hasPrivateDoor
    FROM room_adjacency ra
    JOIN room r ON ra.roomId2 = r.roomID
    WHERE ra.roomId1 = ?
    UNION
    SELECT r.roomNumber, ra.hasPrivateDoor
    FROM room_adjacency ra
    JOIN room r ON ra.roomId1 = r.roomID
    WHERE ra.roomId2 = ?
    """, (room_id, room_id)).fetchall()

conn.close()

return render_template(
    "room_detail.html",
    room=room,
    beds=beds,
    adjacent_rooms=adjacent_rooms
)

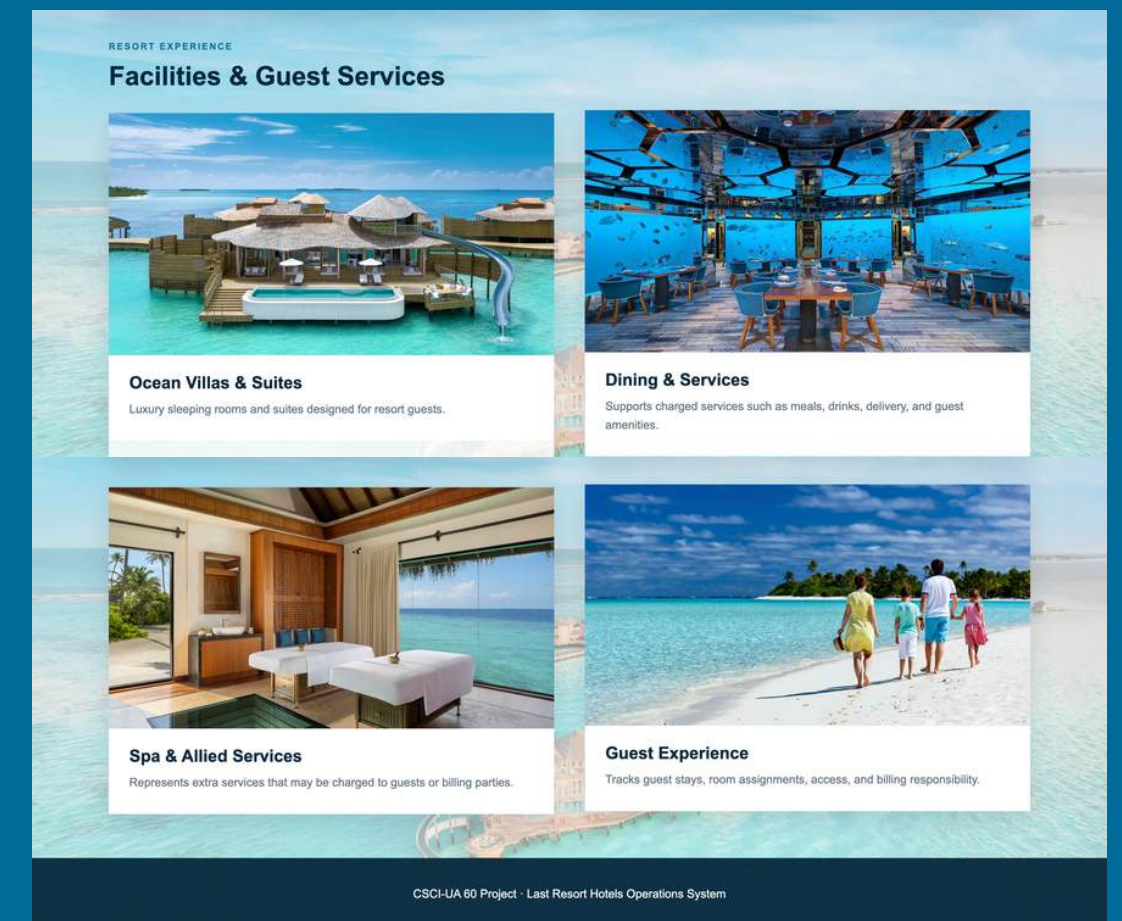
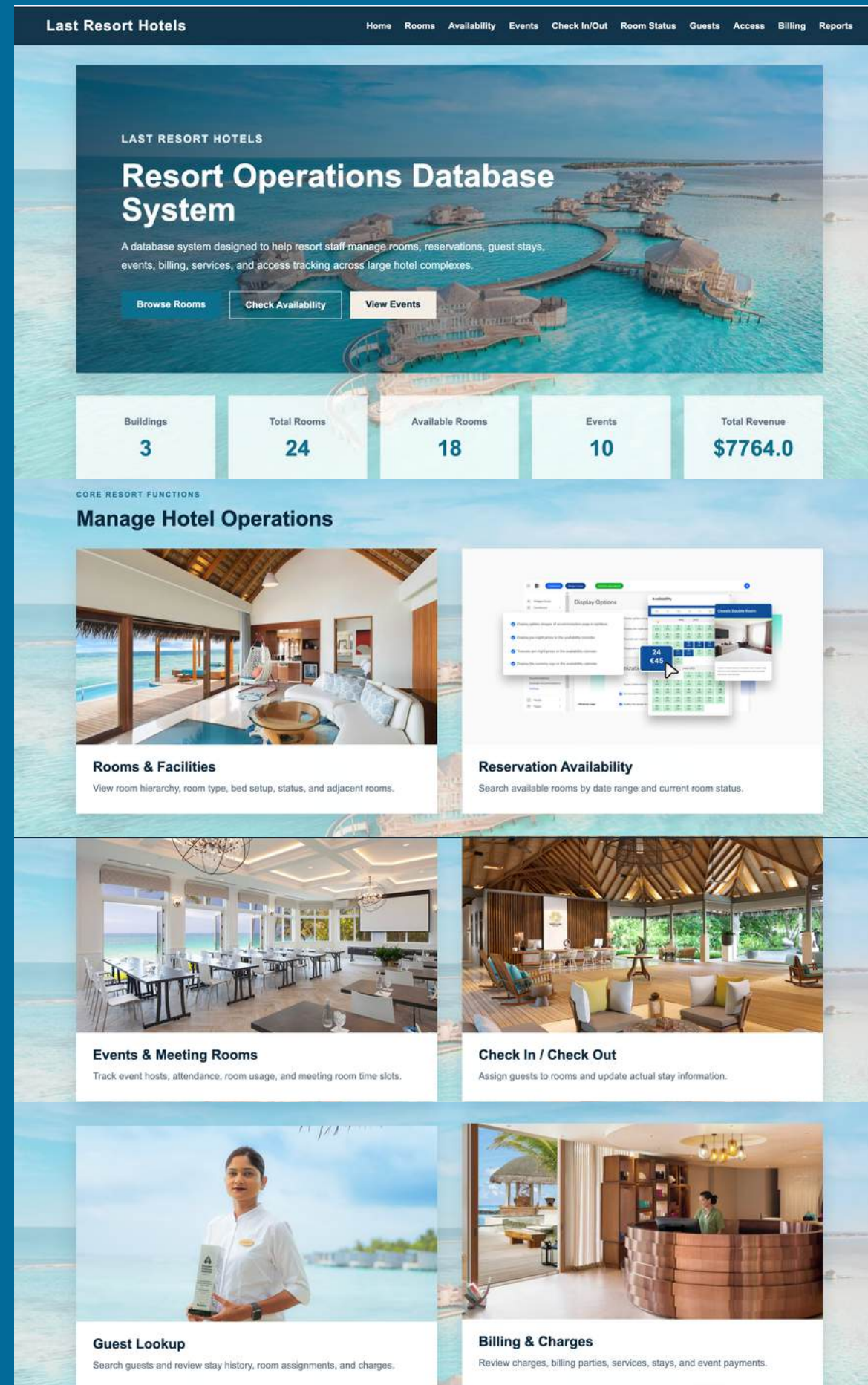
@app.route("/room/<int:room_id>")
def room_detail(room_id):
    conn = get_db_connection()

    room = conn.execute("""
        SELECT r.roomID, r.roomNumber, r.baseRate, r.roomStatus,
               f.floorNumber, f.smokingDesignation,
               w.wingName, w.proximityPool, w.proximityParking, w.handicappedAccess,
               b.buildingName,
               srd.capacity, srd.smoking, srd.hasToilet, srd.hasTV, srd.hasPhone,
               mrd.seatingCapacity
        FROM room r
        JOIN floor f ON r.floorID = f.floorID
        JOIN wing w ON f.wingID = w.wingID
        JOIN building b ON w.buildingID = b.buildingID
        LEFT JOIN sleeping_room_details srd ON r.roomID = srd.roomID
        LEFT JOIN meeting_room_details mrd ON r.roomID = mrd.roomID
        WHERE r.roomID = ?
    """, (room_id,)).fetchone()
```

- Built a Flask web application
- /rooms displays all rooms with full location details
- /room/<id> shows one room's detailed information
- Queries join room, floor, wing, and building tables
- Room detail page includes beds and adjacent rooms
- Flask connects SQL results to HTML templates

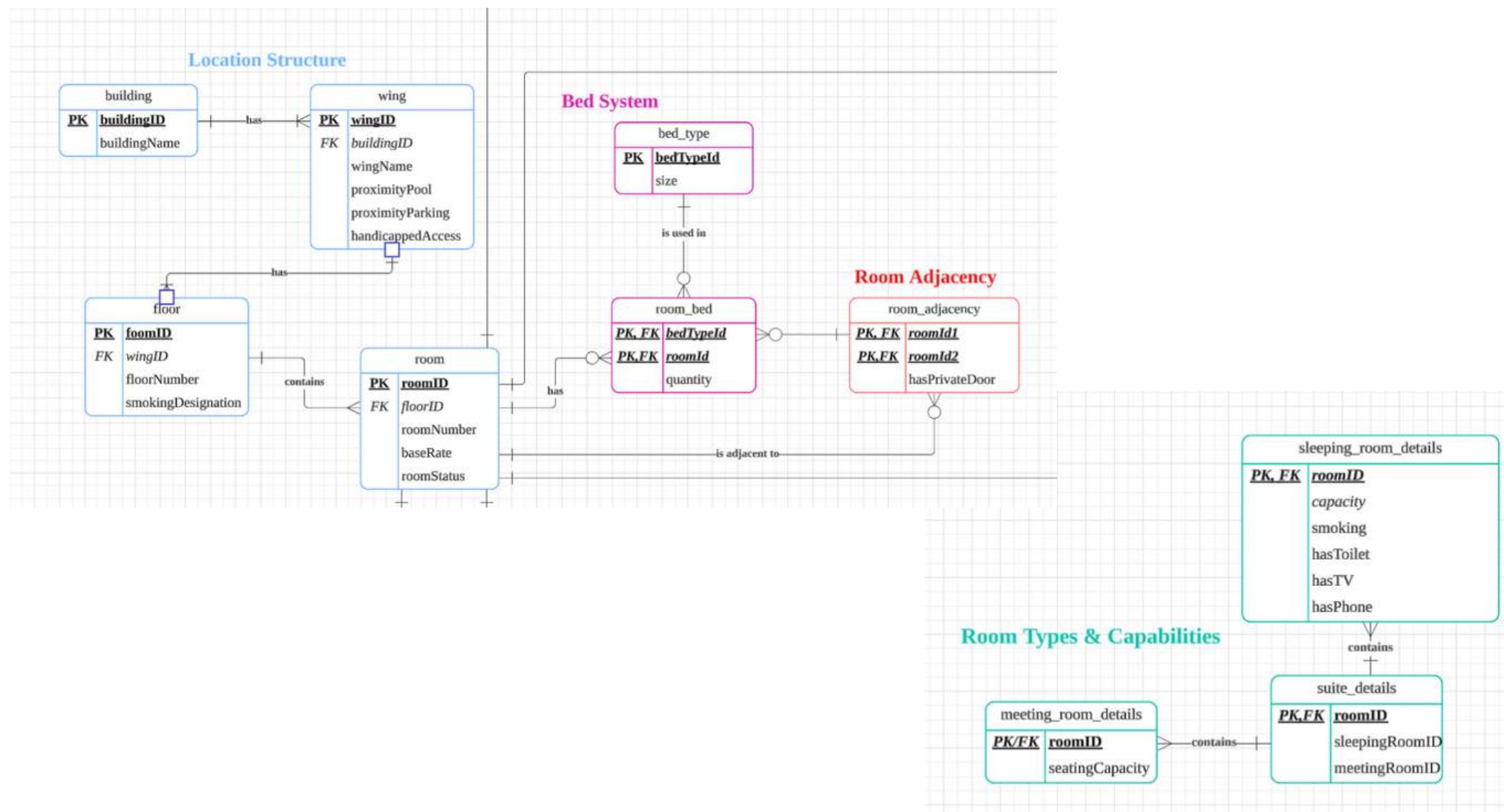
Demo Screens

- Web interface built with Flask
- Homepage summarizes key hotel statistics
- Quick access to room browsing, availability search, and reports
- Displays database results in a user-friendly format
- Connects SQL queries to real hotel operations
- Supports both front-desk tasks and management review



Room & Facility Design

- Rooms belong to floors, wings, and buildings
- Room types include sleeping rooms, suites, meeting rooms, and multi-function rooms
- We track beds, capacity, amenities, room status, and adjacent rooms



Last Resort Hotels Home Rooms Availability Events Check In/Out Room Status Guests Access Billing Reports

ROOM DETAILS

Room N102

Conference Center · Wing North · Floor 1

available Base Rate: \$450.00

Meeting Room

Basic Information

Building: Conference Center
Wing: North
Floor: 1
Base Rate: \$450.00
Status: available
Floor Smoking Designation: non-smoking

Room Function & Features

Sleeping Capacity: N/A
Meeting Seating Capacity: 30
Has Toilet: N/A
Has TV: N/A
Has Phone: N/A

Wing Amenities

Near Pool: No
Near Parking: Yes
Handicapped Access: Yes

Bed Configuration

No bed data available.

Adjacent Rooms

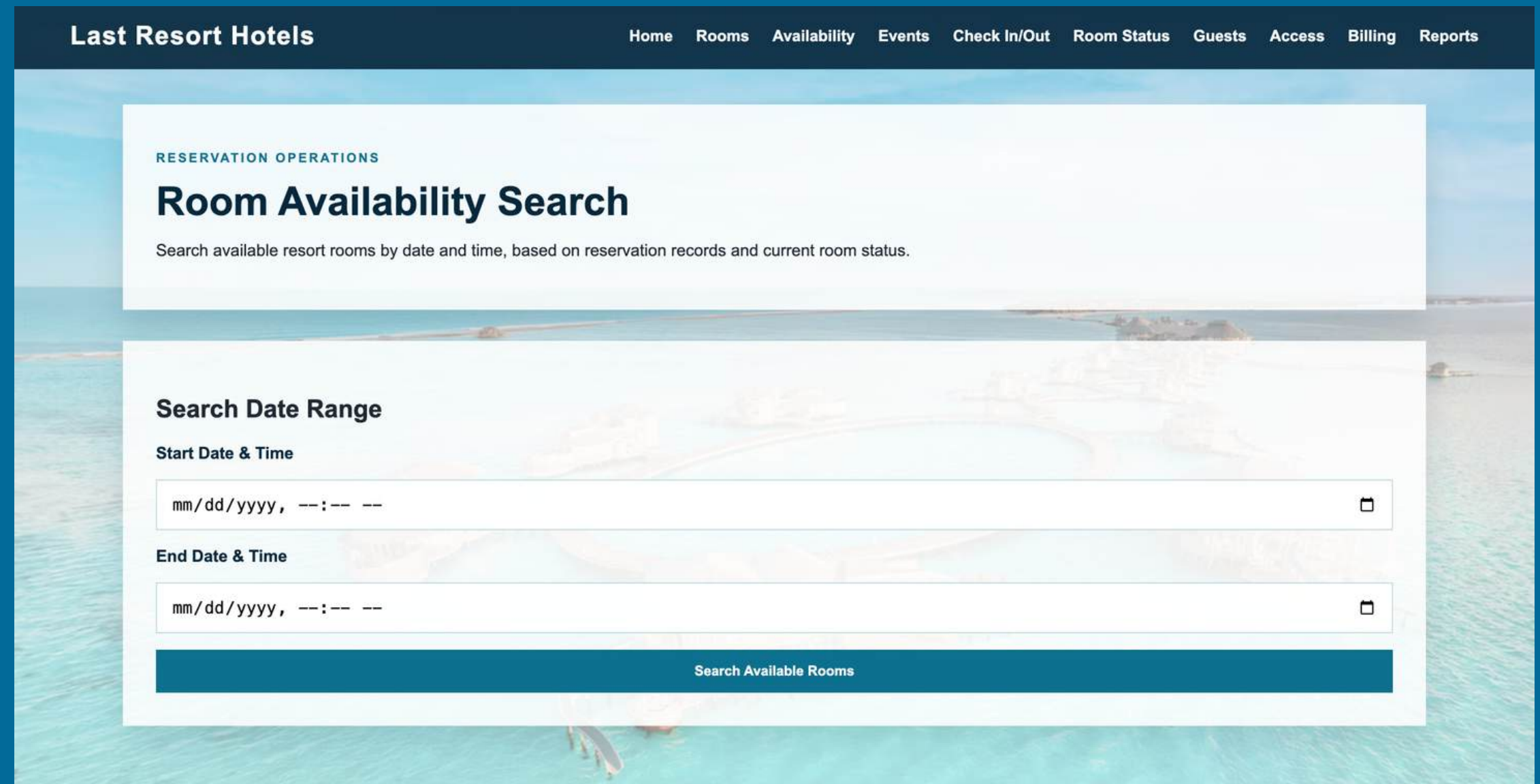
- Room N101 — private-access door
- Room N201 — private-access door

[Back to Rooms](#)

CSCI-UA 60 Project · Last Resort Hotels Operations System

Reservations & Availability

- Reservations can be made for future date ranges
- The web app checks overlapping reservations and current room status to find available rooms



The screenshot displays a web application interface for 'Last Resort Hotels'. At the top, a dark navigation bar contains the site name and a series of menu items: Home, Rooms, Availability, Events, Check In/Out, Room Status, Guests, Access, Billing, and Reports. The main content area features a light blue background with a subtle image of a resort. A white card titled 'RESERVATION OPERATIONS' contains the section header 'Room Availability Search' and a descriptive subtitle. Below this, a 'Search Date Range' section includes two date-time input fields with placeholder text and calendar icons. A dark teal button labeled 'Search Available Rooms' is positioned at the bottom of the form.

Last Resort Hotels

Home Rooms Availability Events Check In/Out Room Status Guests Access Billing Reports

RESERVATION OPERATIONS

Room Availability Search

Search available resort rooms by date and time, based on reservation records and current room status.

Search Date Range

Start Date & Time

mm/dd/yyyy, --:-- --

End Date & Time

mm/dd/yyyy, --:-- --

Search Available Rooms

Guest Stay & Room Assignment

- Guests can check in, be assigned to rooms, and check out
- Room status changes from available → occupied → cleaning

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HomeRoomsAvailabilityEventsCheck In/OutRoom StatusGuestsAccessBillingReports

GUEST OPERATIONS

Check In / Check Out

Manage guest stays, assign rooms, and update room occupancy status.

New Check-In

Guest

Bob Brown · Guest 3

Assigned Room

Room A101 · cleaning

Check-In Time

mm/dd/yyyy, --:-- --

Expected Check-Out Time

mm/dd/yyyy, --:-- --

Check In Guest

Check-Out

Active Stay

Stay 16 · Sophia Tang · Room A101

Check Out Guest

Last Resort Hotels

HomeRoomsAvailabilityEventsCheck In/OutRoom StatusGuestsAccessBillingReports

Current / Recent Stays

Stay ID	Guest	Assigned Room	Check In	Check Out
16	Sophia Tang	Room A101	2026-04-17T15:49	2026-04-28T15:49
15	Noah Lin	Room W101	2026-04-24 14:00	2026-04-26 12:00
14	Mia Huang	Room E101	2026-04-23 14:00	2026-04-27 12:00
13	Leo Yang	Room B201	2026-04-22 14:00	2026-04-24 12:00
13	Leo Yang	Room B202	2026-04-22 14:00	2026-04-24 12:00
12	Karen Wu	Room A201	2026-04-21 14:00	2026-04-22 12:00
11	Jack Chen	Room A101	2026-04-20 14:00	2026-04-23 12:00
11	Jack Chen	Room A102	2026-04-20 14:00	2026-04-23 12:00
10	Ivy Xu	Room W202	2026-04-16 14:00	2026-04-20 12:00
9	Henry Sun	Room W201	2026-04-16 14:00	2026-04-18 12:00

Events & Meeting Rooms

- Events have hosts, date ranges, estimated attendance, estimated guest rooms, and meeting room usage slots
- The Events page supports searching by host/date

Last Resort Hotels

Home

Rooms

Availability

Events

Check In/Out

Room Status

Guests

Access

Billing

Reports

EVENTS MANAGEMENT

Events & Meeting Room Usage

Search events by host or date and review room usage and attendance.

Search Events

Host Name

Enter host name

Date

mm/dd/yyyy

Search

Summary

Total Events: 10

Total Attendance: 460

Estimated Guest Rooms: 73

Last Resort Hotels

Home

Rooms

Availability

Events

Check In/Out

Room Status

Guests

Access

Billing

Reports

Event Details

Event ID	Host	Room	Usage Slot	Start	End	Attendance
1	Doris Yao	Room N101	afternoon	2026-05-01 09:00	2026-05-01 17:00	30
1	Doris Yao	Room N101	morning	2026-05-01 09:00	2026-05-01 17:00	30
2	Ethan Xie	Room N102	lunch	2026-05-02 09:00	2026-05-02 17:00	45
2	Ethan Xie	Room N102	morning	2026-05-02 09:00	2026-05-02 17:00	45
3	Fiona Dong	Room N201	afternoon	2026-05-03 09:00	2026-05-03 17:00	60
3	Fiona Dong	Room N201	morning	2026-05-03 09:00	2026-05-03 17:00	60
4	George Lu	Room S101	morning	2026-05-04 10:00	2026-05-04 18:00	25
5	Hannah Pan	Room S102	afternoon	2026-05-05 08:30	2026-05-05 16:30	40
5	Hannah Pan	Room S102	lunch	2026-05-05 08:30	2026-05-05 16:30	40
6	Ian Yu	Room S201	afternoon	2026-05-06 09:30	2026-05-06 19:00	80

Billing & Charges

- Every charge links to a billing party
- Billing parties can be individual guests or organizations
- Charges can belong to stays, events, rooms, and services

Last Resort Hotels

HomeRoomsAvailabilityEventsCheck In/OutRoom StatusGuestsAccessBillingReports

FINANCIAL OPERATIONS

Billing Review

Search billing parties and review charge ownership, services, stays, events, and responsible payment parties.

Search Billing

Guest Name / Organization ID / Billing Party ID

Example: Grace, 7, or billing party ID

Search Billing Records

Billing Summary

Billing Parties: 17

Total Charges: 35

Total Amount: \$7764.00

Last Resort Hotels

HomeRoomsAvailabilityEventsCheck In/OutRoom StatusGuestsAccessBillingReports

Billing Party Totals		
Billing Party ID	Responsible Party	Total Amount
8	Grace Liu	\$955.00
10	Ivy Xu	\$895.00
17	Organization 7	\$810.00
20	Organization 10	\$700.00
16	Organization 6	\$610.00
7	Frank Zhao	\$570.00
9	Henry Sun	\$560.00
3	Bob Brown	\$551.00
15	Organization 5	\$425.00
4	Cathy Lee	\$355.00
1	John Doe	\$315.00
5	David Kim	\$238.00

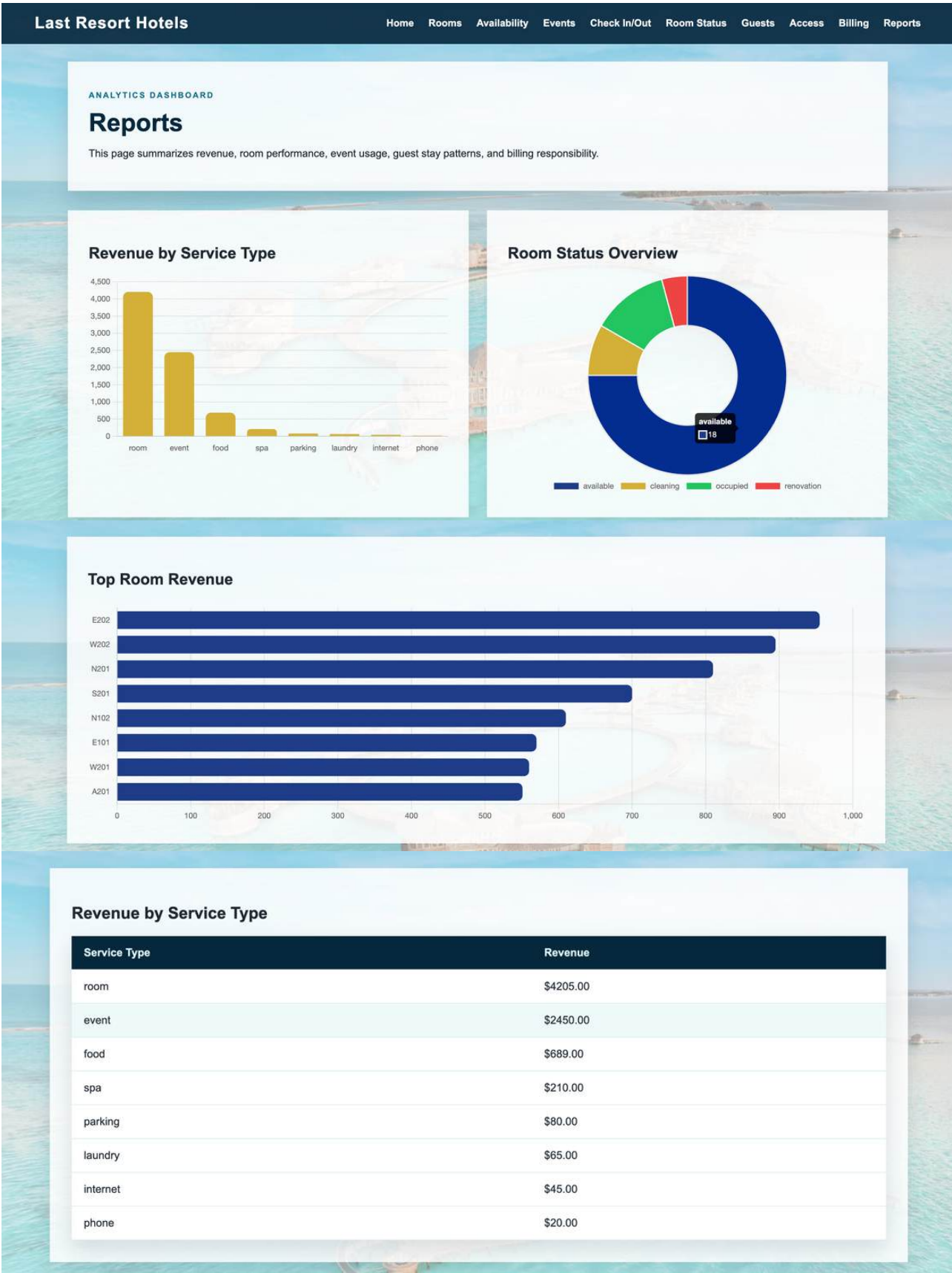
Access Tracking

- Guests use access cards to enter/leave rooms and facilities
- The system records card ID, guest, room, time, and direction

Last Resort Hotels						
Home Rooms Availability Events Check In/Out Room Status Guests Access Billing Reports						
ACCESS TRACKING						
Guest Card Access Logs						
Track guest movement using access cards (enter/exit + room usage).						
Log ID	Guest	Card	Room	Time	Direction	
40	Xavier Jin	25	Room S101	2026-05-10 09:25	enter	
39	Wendy Ren	24	Room N201	2026-05-09 09:20	enter	
38	Victor Cao	23	Room N102	2026-05-08 09:15	enter	
37	Uma Peng	22	Room N101	2026-05-07 09:10	enter	
36	Thomas Li	21	Room S201	2026-05-06 09:05	enter	

35	Sophia Tang	20	Room S102	2026-05-05 09:00	enter	
34	Ryan Ma	19	Room S101	2026-05-04 08:55	enter	
33	Queenie He	18	Room N201	2026-05-03 08:50	enter	
32	Peter Zhang	17	Room N102	2026-05-02 08:45	enter	
31	Olivia Gao	16	Room N101	2026-05-01 08:40	enter	
30	Noah Lin	15	Room W101	2026-04-24 21:30	exit	
29	Noah Lin	15	Room W101	2026-04-24 15:00	enter	
28	Mia Huang	14	Room E101	2026-04-23 20:20	exit	
27	Mia Huang	14	Room E101	2026-04-23 15:00	enter	
26	Leo Yang	13	Room B201	2026-04-22 18:20	exit	
25	Leo Yang	13	Room B201	2026-04-22 15:00	enter	
24	Karen Wu	12	Room A201	2026-04-21 19:30	exit	

Reports & Analytics



Revenue by Room

Room Number	Total Revenue
E202	\$955.00
W202	\$895.00
N201	\$810.00
S201	\$700.00
N102	\$610.00
E101	\$570.00
W201	\$560.00
A201	\$551.00
N101	\$425.00
A203	\$355.00
A101	\$315.00
B101	\$238.00

Guests with Multiple Stays

No guests currently have multiple stays in the sample data.

Event Schedule and Rooms Used

Event ID	Start	End	Room	Usage Slot
1	2026-05-01 09:00	2026-05-01 17:00	N101	afternoon
1	2026-05-01 09:00	2026-05-01 17:00	N101	morning
2	2026-05-02 09:00	2026-05-02 17:00	N102	lunch
2	2026-05-02 09:00	2026-05-02 17:00	N102	morning
3	2026-05-03 09:00	2026-05-03 17:00	N201	afternoon
3	2026-05-03 09:00	2026-05-03 17:00	N201	morning
4	2026-05-04 10:00	2026-05-04 18:00	S101	morning
5	2026-05-05 08:30	2026-05-05 16:30	S102	afternoon

Charge Ownership and Billing Responsibility

Charge ID	Service	Amount	Charge Time	Room	Belongs To	Reference	Billing Party	Description
1	room	\$240.00	2026-04-14 12:00	A101	Stay	Stay 1	John Doe	Room charge for stay 1
2	room	\$120.00	2026-04-13 12:00	A102	Stay	Stay 2	Alice Smith	Room charge for stay 2
3	room	\$450.00	2026-04-16 12:00	A201	Stay	Stay 3	Bob Brown	Room charge for stay 3
4	room	\$330.00	2026-04-15 12:00	A203	Stay	Stay 4	Cathy Lee	Room charge for stay 4
5	room	\$220.00	2026-04-16 12:00	B101	Stay	Stay 5	David Kim	Room charge for stay 5
6	room	\$145.00	2026-04-15 12:00	B201	Stay	Stay 6	Emma Wang	Room charge for stay 6
7	room	\$540.00	2026-04-18 12:00	E101	Stay	Stay 7	Frank Zhao	Room charge for stay 7
8	room	\$840.00	2026-04-19 12:00	E202	Stay	Stay 8	Grace Liu	Room charge for stay 8
9	room	\$440.00	2026-04-18 12:00	W201	Stay	Stay 9	Henry Sun	Room charge for stay 9

Reports answer business questions: revenue by service type, revenue by room, room status distribution, guests with multiple stays, event schedules, and billing ownership



Thank You!